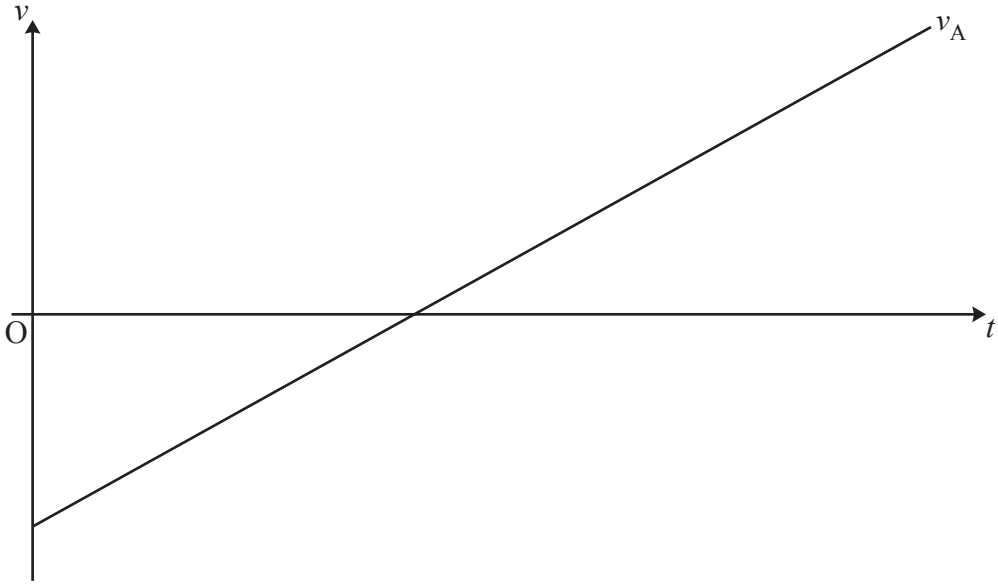
13 Particles A and B are moving along the same straight line. The graph below is the velocity–time graph for particle A with velocity $v_A \text{ m s}^{-1}$ at time t s. Particle A travels with constant acceleration. It has an initial velocity of -3 m s^{-1} and is instantaneously at rest when $t = 4$.



The velocity for particle B at time t s is given by $v_B = 1.5 + 0.03t^2 \text{ m s}^{-1}$. The particles are **first** travelling at the same velocity at time $t = T$ s. You are given that A and B do not collide.

(a) Show that $T = 10$. **[3]**

(b) Hence, or otherwise, determine which particle has travelled the greater **distance** by time T s and by how much. **[6]**